

HELLINA HAILU NIGATU

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[Website](#)

I am interested in improving access and utility of language technologies by (1) centering linguistic characteristics and community collaboration in design and (2) investigating the impact of currently deployed language technologies on marginalized communities in the Global Majority. My research area lies in the intersection of NLP, HCI, and AI Ethics.

EDUCATION

PhD in Computer Science, University of California at Berkeley. Dec 2025
Advised by [Sarah Chasins](#) and [John Canny](#) CGPA: 3.78 [Thesis](#)

MSc in Computer Science, University of California at Berkeley. May 2023
Advised by [Sarah Chasins](#) and [John Canny](#) CGPA: 3.73

BSc. Electrical and Computer Engineering, Addis Ababa University Dec 2020
Advised by [Biruk Tilahun](#) and [Feven Fisseha](#) CGPA: 3.89

PUBLICATIONS

Yeswa-Stories: A Three-Way Parallel Dataset of Female African Figures in Low-Web Data Languages GITT 2026
Hellina Hailu Nigatu, Bethelhem Yemane Mamo
In Proceedings of the Workshop on Gender Inclusive Translation Technologies.

Evaluating Machine Translation Datasets for Low-Web Data Languages: A Gendered Lens. ACL 2026
Hellina Hailu Nigatu, Bethelhem Yemane Mamo, Bontu Fufa Balcha, Debora Taye Tesfaye, Elbethel Daniel Zewdie, Ikram Behiru Nesiru, Jitu Ewnetu Hailu, Senait Mengesha Yayo
In Proceedings of the Findings of Association of Computational Linguistics Conference.

UnWEIRDing Peer Review in Human-Computer Interaction CHI 2026
Hellina Hailu Nigatu, Farhana Shahid, Vishal Sharma, Abigail Oppong, Michaelanne Thomas, Syed Ishtiaque Ahmed
In Proceedings of the Computer-Human Interactions conference.

Viability of Machine Translation for Healthcare in Low-Resourced Languages EMNLP 2025
Hellina Hailu Nigatu, Nikita Mehandru, Negasi Haile Abadi, Blen Gebremeskel, Ahmed Alaa, Monojit Choudhury
In Proceedings of the Empirical Methods for Natural Language Processing.

A Case Against Implicit Standards: Homophone Normalization in Machine Translation for Languages that use the Ge'ez Script. EMNLP 2025
Hellina Hailu Nigatu, Atnafu Lambebo Tonja, Henok Biadgign Ademtew, Hizkiel Mitiku Alemayehu, Negasi Haile Abadi, Tadesse Destaw Belay, Seid Muhie Yimam
In Proceedings of the Empirical Methods for Natural Language Processing.

Into the Void: Understanding Online Health Information in Low-Web Data Languages AIES 2025
Hellina Hailu Nigatu, Nuredin Ali Abdelkadir, Fiker Tewelde, Stevie Chancellor, Daricia Wilkinson
In Proceedings of the Artificial Intelligence, Ethics, and Society (Upcoming).

mRAKL: Multilingual Retrieval-Augmented Knowledge Graph Construction for Low-Resourced Languages. ACL (Findings) 2025
Hellina Hailu Nigatu, Min Li, Maartje ter Hoeve, Saloni Potdar, Sarah E. Chasins
In Proceedings of Findings of the Association of Computational Linguistics.

Exploring-Transliteration-Based-Zero-Shot-Transfer-for-Amharic-ASR AfricaNLP 2025
Hellina Hailu Nigatu, Hanan Aldarmaki
In Proceedings of the AfricaNLP Workshop.

Cognate Detection for Historical Language Reconstruction of Proto-Sabean Languages: the Case of Ge'ez, Tigrinya, and Amharic COLING 2025

Elleni Sisay Temesgen, Hellina Hailu Nigatu, and Fitsum Assamnew Andargie.

In Proceedings of the 31st International Conference on Computational Linguistics.

The Zeno's Paradox of Low-Resource Languages. EMNLP 2024

Hellina Hailu Nigatu, Atnafu Lambebo Tonja, Benjamin Rosman, Thamar Solorio, Monojit Choudhury

In Proceedings of the Empirical Methods for Natural Language Processing. –**Outstanding Paper Award**

Exploitation All the Way Down: Calling out the Root Cause of Bad Online Experiences for Users of the Majority World. DC-DAIG 2024

Hellina Hailu Nigatu, Zeerak Talat.

Data and AI Governance Coalition (DAIG) at the 19th Annual Meeting of Internet Governance Forum.

A Capabilities Approach to Studying Bias and Harm in Language Technologies. WAI 2024

Hellina Hailu Nigatu, Zeerak Talat.

Extended Abstract accepted to Workshop on New Perspectives on Bias and Discrimination in Language Technology.

Evaluating Gender Bias in Machine Translation for Low-Resource Languages AfricaNLP 2024

Walelign Tewabe Sewunetie, Atnafu Lambebo Tonja, Tadesse Destaw Belay, Hellina Hailu Nigatu, Gashaw Kidanu,

Zewdie Mossie, Hussien Seid, Eshete Derb, Seid Muhie Yimam

Proceedings of the 5th Workshop on African Natural Language Processing

“I Searched for a Religious Song in Amharic and Got Sexual Content Instead.”: Investigating Online Harm in Low-Resourced Languages on YouTube. FAccT 2024

Hellina Hailu Nigatu, Inioluwa Deborah Raji

In Proceedings of the ACM Conference on Fairness, Accountability, and Transparency

Low-Resourced Languages and Online Knowledge Repositories: A Need Finding Study. CHI 2024

Hellina Hailu Nigatu, John Canny, Sarah Chasins

In Proceedings of the CHI Conference on Human Factors in Computing Systems

The Less the Merrier? Investigating Language Representation in Multilingual Models EMNLP 2023

Hellina Hailu Nigatu, Atnafu Lambebo Tonja, Jugal Kalita

In Proceedings of Empirical Methods in Natural Language Processing (EMNLP) 2023 Findings

Co-Designing for Transparency: Lessons from Building a Document Organization Tool for the Criminal Justice Domain. FAccT 2023

Hellina Hailu Nigatu, Lisa Pickoff-White, John Canny, Sarah Chasins

In Proceedings of ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT)

A Need Finding Study with Low-Resource Language Content Creators AfriCHI 2023

Hellina Hailu Nigatu, John Canny, Sarah Chasins

In Proceedings of 4th ACM African Human-Computer Interaction Conference (AfriCHI 2023)

Enhancing Translation for Indigenous Languages: Experiments with Multilingual Models. AmericasNLP 2023

Atnafu Lambebo Tonja, Hellina Hailu Nigatu, Olga Kolesnikova, Grigori Sidorov, Alexander Gelbukh, Jugal Kalita

In AmericasNLP 2023 Shared Task on Machine Translation into Indigenous Languages

DyGraph: A Dynamic Graph Generator and Benchmark Suite GRADES-NDA 2022

Andrew McCrabb, Hellina Nigatu, Absalat Getachew and Valeria Bertacco

In Proceedings of ACM Workshop for Graph Data Management Experiences and Systems and Network Data Analytics

The African Stopwords Project: Curating Stopwords for African Languages AfricaNLP 2022

Chris Chinenye Emezue, Hellina Hailu Nigatu, Cynthia Thinwa, Lerato Louis, Idris Abdulmumin, Samuel Gbenga Oy-erinde, Benjamin Ayoade Ajibade, Helper Zhou, Emeka Felix Onwuegbuzia, Handel Chiagozie Emezue, Ifeoluwatayo Adeseye Ige, Atnafu Lambebo Tonja, Chiamaka Ijeoma Chukwuneke, Shamsuddeen Hassan Muhammad, Olanrewaju Samuel.

In Proceedings of ICLR Workshop for African Natural Language Processing.

PROJECTS

Multilingual Knowledge Graph Completion

- Collaboration with Apple MLR and Product team.
- Focusing on Amharic, Tigrinya, and Arabic.

Improving Automatic Speech Recognition (ASR) for Amharic.

- Based on findings from need-finding work, working on improving ASR for Amharic.
- Investigating transfer learning approaches and impact of transfer language selection in system improvement.

Understanding Multilingual Language Models

[Paper](#)

- Conducted experiments to understand the differences in embedding space representations across languages in Multilingual Language Models.
- Compared community-based (models built for specific language families or geographic locations and by communities who speak those languages) to generic models (models that include several languages with no clear relation).
- Compared representation among language families, writing scripts, and dialects.
- Investigated how different language representations relate to performance in models being able to distinguish between languages downstream.

Auditing YouTube in Low-Resourced Languages

- Auditing YouTube for online harm in Low-Resourced Languages, particularly Amharic.
- Used HCI methods for quantitative and qualitative analysis of data.

Need-finding Study of Low-Resourced Language Content Creators

- Conducted two separate studies to understand the challenges—technological and societal—that prevent low-resourced language content creators interacting with online knowledge repositories.
- Used Contextual Inquiry and qualitative content analysis for the study.

AI Augmented Document Organization Tool (Masters Thesis)

[GitHub Paper](#)

- Building a tool for social scientists to organize and sort through large amounts of messy data based on specific entities within the data.
- Pooling techniques from NLP, ComVis, image processing, and HCI.

DyGraph: A Dynamic Graph Generator

[GitHub Poster](#)

- Implemented a dynamic graph generator in C++ that outputs graphs that mimic properties of real-life dynamic graphs such as social networks.
- Focused on capturing the power law nature of degree distributions observed in the naturally existing dynamic graphs.

Obfuscating Electric Circuits by Replicating Logic Gates

[Poster](#)

- Implemented a tool to replicate sets of connected gates by taking in a Verilog description of a hardware and building a data structure of the netlist in the description.
- Objective of the tool was to protect logical circuits from hardware trojans by adding obfuscation using replicated gates and crossbars.

Analyzing CNN news transcripts to identify frequency of interruptions

[GitHub](#)

- A mini-project that was mainly to serve as an NLP learning experience.
- Used NLP techniques like Sentiment Analysis and Topic Modeling to analyze the characteristics of each of the selected anchors and shows.

- This project arose as the need to mitigate the challenges of access to healthcare facilities in Ethiopia.
- A Google Maps integrated mobile application that allows patients to search for healthcare services on their phones and locate the nearest facilities that provide those services.
- A web application where facilities enter the products and services they have so patients can access them.
- The metadata for the project is generated using a dynamic graph generator I built for my research project in order to match real-life demands of patients.

AWARDS AND HONORS

- **Wangari Maathai Impact Award 2026**
- **Wikimedia Foundation Research of the Year Award 2025**
- **Outstanding Paper Award, EMNLP 2024**
- **Fineunning Meta Models on African Languages Hackaton(3rd place), Deep Learning Indaba 2024**
- **Best Poster Award, Deep Learning Indaba 2024**
- **Data Science for Health Ideathon (2nd place), Deep Learning Indaba 2024**
- **FAccT DEI Scholar, ACM FAccT**
- **ACM-W Scholarship for Attendance of Research Conferences program, ACM**
- **SIGHPC Computational and Data Science Fellowship, SIGHPC**
- **BAIR Research Ignition Award, UC Berkeley, Berkeley Artificial Intelligence Research Lab**
- **EECS Excellence Award, UC Berkeley, Department of Electrical Engineering and Computer Sciences**
- **Award for Highest Cumulative GPA, Women 1st Place, Addis Ababa Institute of Technology**
- **Award for Highest Cumulative GPA, Women 1st Place, Addis Ababa Institute of Technology, Department of Electrical and Computer Engineering**
- **BSc with Very Great Distinction, Addis Ababa Institute of Technology, Department of Electrical and Computer Engineering**

EXPERIENCE

Postdoc

AI Accountability Lab

Current

Dublin, Ireland

- Collaborating with a community organization in Mali who are building a speech recognition system for their language. Focused on data governance and release, model development, and evaluation.

Research Intern

Distributed AI Research (DAIR) Institute

May 2025 - August 2025

Remote

- Worked as a summer intern on language technology support for Wikipedia article creation in dialects of low-resourced languages.

Research Assistant

Mohammed Bin Zayed University of Artificial Intelligence

June 2024 - August 2024

Abu Dhabi, UAE

- Working as a summer intern on Machine Translation in Health Care in Prof. Monojit Choudhury's Lab.

Data Scientist

Opian Information Communication Technology

Jan 2021 - May 2021

Addis Ababa, Ethiopia

- Served as a team lead for the Data Science division of the company.
- Led projects related to utilizing police report data to assess reporting climate in relation to Gender Based Violence cases and analyzing tourist review data to extract insights about Ethiopia's tourism sector.

- Continually conduct requirement analysis for other projects in the company and suggest improvements and fixes for the existing tools.

Research Assistant
University of Michigan

Aug 2019 - April 2020
Addis Ababa, Ethiopia (Remote role)

- As an extension of prior working engagement with the University, I improved and added features to projects I had implemented during my position as an undergraduate research assistant in Summer 2019.

Data Scientist
Opian Information Communication Technology

May 2020 - Oct 2020
Addis Ababa, Ethiopia

- Initially worked as a researcher and student mentee under the student mentorship program. Was promoted to full time data science researcher and analyst.
- Undertook research for projects related to Machine Learning for Healthcare; specifically, carried out tasks to improve pre-existing prediction tools of the company.
- After proposing a new project, which was approved by my supervisor, I worked on adopting a machine learning prediction tool for Intensive Care Units (ICUs) to the Ethiopian context.

Undergrad Research Assistant
University of Michigan

May 2019 - Aug 2019
Ann Arbor, MI

- African Undergraduate Research Adventure (AURA) participant:
- Executed research for projects related to graph computing and hardware security.
- Developed tools in support of the research projects.

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